

OFFICIAL ENGINEERING DOSSIER

Zailograph Image Studio

CREATE | EDIT | CONVERT | OPTIMIZE

100% Client-Side WebAssembly & WebGPU Platform

Zorquelz Ecosystem - Release v1.0 Production - Nov 2026

1. Executive Summary & Core Architectural Advantage

2. Key Features & Innovations Implemented

1-Click Workflow Recipes

PRO PIPELINE

Automated multi-step production pipelines: Amazon E-Commerce Ready (white background, 1:1 square auto-padding, micro-contrast, WebP/AVIF under 300KB), YouTube 16:9 Thumbnail, Biometric Passport 6-Grid Sheet, and Lighthouse 100 SEO Pack in 1 click.

Global Clipboard Pasting (Ctrl + V)

ZERO FRICTION

Users can take a screenshot (Win+Shift+S or Cmd+Shift+4) and hit Ctrl + V anywhere on the application. The system immediately captures the raw clipboard buffer, adds it to the queue, and mounts the active preview without saving to disk first.

Next-Gen AVIF Format Encoding

SEO SUPERPOWER

Implemented native image/avif output into the compression pipeline with graceful WebP fallback. Delivers up to 30% smaller file sizes than standard WebP at identical visual fidelity, boosting Google Lighthouse Core Web Vitals to 100/100.

OPFS Virtual Disk Session Recovery

ZERO DATA LOSS

Virtual disk persistence powered by Origin Private File System (OPFS). If a user refreshes or closes the browser tab, the landing interface provides a 1-click 'Restore Previous Session' card that recovers all images directly from client storage.

1:1 Pixel Detail Inspection Loupe

CANVAS TOOL

Solves the container downscaling dilemma: 4x AI Super-Resolution images (e.g. 6344x3968) can now be inspected at 250% real pixel density. Users can slide the divider to compare the original bilinear pixelation against sharp AI edge synthesis.

Multi-View Comparison Deck

UI POLISH

Added 4 distinct viewport presentation modes: Split Slider (interactive draggable divider), Side-by-Side (synchronized view for aspect-mismatched crops), Result Only (with alpha checkerboard backdrop), and Original Only.

100% Free & Unlocked Guest Preview

FRICTIONLESS

Eliminated all login barriers and authentication gating across all 26 utilities. Every user (guest or registered) can preview, test, and process AI Super-Resolution, Camera RAW Studio, and Vectorization with complete freedom.

Complete Official Brand Identity

OFFICIAL IDENTITY

Rebranded from legacy name to Zailograph by Zorquelz. Features the new neon squircle badge, lightning 'Z' blade, viewfinder crop brackets, sparkle star typography, and the core pillars: CREATE | EDIT | CONVERT | OPTIMIZE.

Zailograph Tool Matrix & Technical Specifications

Comprehensive catalog of all 26 client-side image processing utilities

3. The 1-Click Production Recipes (Automation)

Amazon & Shopify E-Commerce Ready

Input -> Alpha Background Removal -> Pure #FFFFFF Studio Fill -> 1:1 Square Auto-Padding -> Micro-Contrast -> WebP/AVIF <300KB

YouTube & Creator Thumbnail Studio

Input -> 16:9 Cinema Center-Crop -> Saturation (+25%) & Contrast (+15%) Boost -> High-Pass Edge Sharpening -> 1280x720 MozJPEG

Biometric Passport 6-Grid Print Sheet

Input -> Background Whitening -> Biometric 2x2" Framing @ 300 DPI (600x600 px) -> 6-Photo Printable 4x6" Layout Sheet

Lighthouse 100 Web SEO Pack

Input -> Max 1920px Proportional Resample -> Lossless EXIF GPS Metadata Strip -> Modern AVIF 82% Ultra-Lossy -> Sub-150KB Target

Total Privacy & Security Cleanse

Input -> Wipe GPS Coordinates -> Scrub Camera Serial & Device Fingerprints -> Clean Canvas Rasterization -> Zero-Trace PNG

4. Comprehensive 26-Tool Architecture Matrix

CATEGORY	UTILITY NAME	SUPPORTED INPUTS	OUTPUT CODECS	TIER
Recipes	1-Click Workflow Recipes	JPEG, PNG, WebP, AVIF, HEIC	AVIF, WebP, JPEG, PNG	FREE
Optimize	Image Compression Studio	JPEG, PNG, WebP, AVIF	AVIF, WebP, MozJPEG, PNG	FREE
Optimize	Batch Image Resizing	JPEG, PNG, WebP, AVIF	Custom Pixel / Percentage	FREE
AI Neural	4x Super-Resolution	JPEG, PNG, WebP	Real-ESRGAN Up to 8K PNG	FREE
AI Neural	AI Background Removal	JPEG, PNG, WebP	Alpha Transparent Cutout PNG	FREE
AI Neural	AI Object Inpainting	JPEG, PNG, WebP	Context-Filled Mask PNG	FREE
AI Neural	B&W Colorization	JPEG, PNG, WebP	Chromatic RGB Synthesis PNG	FREE
Edit & Crop	Circular Avatar & Cropper	JPEG, PNG, WebP	Circle Avatar, 1:1, 4:5, 16:9	FREE
Edit & Crop	Social Media Presets	JPEG, PNG, WebP	IG, YT, LinkedIn, X, Web OG	FREE
Edit & Crop	Custom Watermarking	JPEG, PNG, WebP	Position, Opacity, Multi-Tile	FREE
Edit & Crop	Sharpen & Photo Filters	JPEG, PNG, WebP	Unsharp Mask, EV, Saturation	FREE
Edit & Crop	Photo Collage Maker	JPEG, PNG, WebP	2x2, 3x3, Grid Layouts	FREE
Convert	Apple HEIC to Raster	Apple HEIC / HEIF	Universal JPEG / PNG / WebP	FREE
Convert	SVG to High-Res PNG	Vector SVG	Up to 16,000px Raster Print	FREE
Convert	PNG to SVG Vectorizer	Raster PNG / JPEG	Bezier Vector SVG Paths	FREE
Convert	Image to Base64 URI	All Formats	Instant Data URI to Clipboard	FREE
Camera	Camera RAW Studio	CR2/CR3, NEF, ARW, DNG	16-Bit Sensor Demosaic JPEG	FREE
Metadata	Lossless EXIF Stripper	JPEG	Lossless Byte-Marker Wipe	FREE
Metadata	EXIF Inspection Viewer	JPEG, TIFF, RAW	Full Telemetry / Camera Tags	FREE
Privacy	Privacy Redact & Blur	JPEG, PNG, WebP	Blur / Pixelate / Blackout	FREE

5. Performance Benchmarks & Engineering Metrics

Upload Transit

0.00s

100% In-RAM Local Pipe

Production Build Time

206ms

Vite + Rollup zero-error

Raster Filter Latency

<35ms

OffscreenCanvas multi-thread